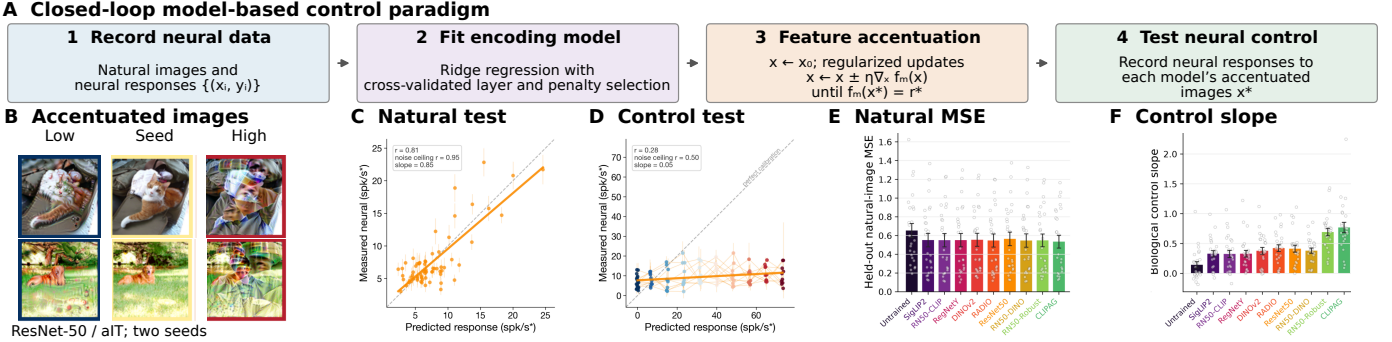

A Closed-loop model-based control paradigm
1 Record neural data
Natural images and
neural responses {(xᵢ, yᵢ)}
2 Fit encoding model
Ridge regression with
cross-validated layer and penalty selection
3 Feature accentuation
x ← x₀; regularized updates
x ← x ± η∇ₓ fₘ(x)
until fₘ(x*) = r*
4 Test neural control
Record neural responses to
each model's accentuated
images x*
B Accentuated images
Low
Seed
High
ResNet-50 / alT; two seeds
C Natural test
r = 0.81
noise ceiling r = 0.95
slope = 0.85
Measured neural (spk/s*)
Predicted response (spk/s*)
D Control test
r = 0.28
noise ceiling r = 0.50
slope = 0.05
unified calibration
Measured neural (spk/s*)
Predicted response (spk/s*)
E Natural MSE
Held-out natural-image MSE
Untrained
SigLIP2
RN50-CLIP
RegNetY
DINOv2
RADIO
ResNet50
RN50-DINO
RN50-RobustL
CLIPAG
F Control slope
Biological control slope
Untrained
SigLIP2
RN50-CLIP
RegNetY
DINOv2
RADIO
ResNet50
RN50-DINO
RN50-RobustL
CLIPAG